

Notes: Hansen & Singleton (JPE, 1983)

Stochastic Consumption, Risk Aversion, and the Temporal Behavior of Asset Returns

Contents

1	Big picture	3
2	Model and measurement (key objects)	3
2.1	Representative-agent preferences	4
2.2	Asset-pricing Euler equation	4
2.3	Key transformed variables	4
2.4	Distributional assumption	5
2.5	What the paper does and does not model explicitly	5
2.6	Consumption data	5
2.7	Return data	5
2.8	Timing assumption behind monthly data	6
3	Models and empirical specifications (all equations + interpretation)	6
3.1	The main log-linear restriction	6
3.2	Special cases	7
3.3	The model's predictability formula	7
3.4	Forecasting equation for consumption growth	7
3.5	Restricted VAR representation	8
3.6	Maximum-likelihood estimation	8
3.7	Lag length and quarterly comparison	8
3.8	A second class of tests using only nominal returns	9
4	Tables and empirical results	9
4.1	Table 1: value-weighted aggregate stock return	9
4.2	Table 2: why increasing lag length does not change much	10
4.3	Table 3: individual Dow Jones returns	11
4.4	Table 4: 1-month Treasury bill return	11
4.5	Table 5: stock and bond returns jointly	12
4.6	Nominal-return-spread tests	13
5	Identification and what is really being tested	13
5.1	Where identification of α and β comes from	13

5.2	What a rejection means	13
5.3	Why the aggregate stock return looks better than individual stocks	14
5.4	Timing and measurement concerns	14
5.5	How to read the α estimates economically	14
5.6	Why the Treasury-bill rejection matters	15
6	Short summary	15
7	Conclusion	15
7.1	Core conclusion	15
7.2	Economic intuition	16
7.3	Takeaway	16

1 Big picture

- **Question.** What restrictions does a representative-agent asset-pricing model impose on the *joint time-series behavior* of aggregate consumption and asset returns?
- **Setting.** A representative consumer has time-separable CRRA preferences and trades assets to smooth consumption over time. The paper studies both stocks and short-term Treasury bills.
- **Main conceptual move.** Under two strong assumptions — **CRRA utility** and a **joint stationary Gaussian / lognormal law** for consumption growth and returns — the Euler equations become a *restricted log-linear forecasting system* for log returns and log consumption growth.
- **Key theoretical result.** The predictable component of each asset's log return must be proportional to the predictable component of log consumption growth. The common proportionality factor is **minus the coefficient of relative risk aversion**.
- **Special implication.** If agents are risk neutral, log returns should be serially uncorrelated. More generally, predictable returns require both **risk aversion** and **forecastable consumption growth**.
- **Empirical strategy.** Estimate the preference parameters and the consumption-forecasting process by maximum likelihood, then test the overidentifying restrictions implied by the model.
- **Main data.** Monthly postwar U.S. data, **1959:2–1978:12**, using real per-capita consumption of either **nondurables (ND)** or **nondurables plus services (NDS)**, together with stock returns and Treasury bill returns.
- **Main aggregate-stock result.** The restrictions are **not strongly rejected** when the return is the value-weighted NYSE stock return. The estimates of risk aversion are economically plausible but very imprecise.
- **Main rejection result.** The model is **strongly rejected** for individual Dow Jones stocks, for 1-month Treasury bills, and for additional tests based only on nominal return spreads.
- **Big empirical lesson.** The representative-agent CRRA–lognormal structure can fit some *aggregate* return behavior, but it breaks down much more sharply once one looks at richer cross-asset evidence or tests that avoid consumption measurement.
- **Historical importance.** This paper is an early attempt to bring modern consumption-based asset pricing to time-series data using explicit likelihood methods. It sits right before the later Hansen–Singleton move toward distribution-free GMM methods.

2 Model and measurement (key objects)

2.1 Representative-agent preferences

The representative consumer has period utility

$$U(c_t) = \frac{c_t^\gamma}{\gamma}, \quad \gamma < 1, \quad (1)$$

and maximizes

$$E_0 \sum_{t=0}^{\infty} \beta^t U(c_t), \quad 0 < \beta < 1. \quad (2)$$

Interpretation. The coefficient of relative risk aversion is

$$1 - \gamma.$$

The paper rewrites this as

$$\alpha \equiv \gamma - 1 < 0,$$

so that the coefficient of relative risk aversion is $-\alpha$. Thus:

- $\alpha = 0$ means risk neutrality,
- $\alpha = -1$ means logarithmic utility,
- more negative α means greater risk aversion.

2.2 Asset-pricing Euler equation

Let $r_{i,t+1}$ be the gross real return on asset i . The first-order condition is

$$E_t \left[\beta \left(\frac{c_{t+1}}{c_t} \right)^\alpha r_{i,t+1} \right] = 1, \quad i = 1, \dots, N. \quad (3)$$

Interpretation. Expected returns are restricted by how asset payoffs covary with the marginal utility of consumption. Because $\alpha < 0$, bad consumption states receive high marginal-utility weight.

2.3 Key transformed variables

The paper defines

$$x_t \equiv \frac{c_t}{c_{t-1}}, \quad X_t \equiv \log x_t, \quad R_{it} \equiv \log r_{it}. \quad (4)$$

It also defines

$$u_{it} \equiv x_t^\alpha r_{it}, \quad U_{it} \equiv \log u_{it} = \alpha X_t + R_{it}. \quad (5)$$

Let

$$Y_t \equiv (X_t, R_{1t}, \dots, R_{nt})'. \quad (6)$$

Interpretation. The empirical system is built from **log consumption growth** and **log asset returns**. The Euler equation becomes especially tractable once everything is written in logs.

2.4 Distributional assumption

The central assumption is that $\{Y_t\}$ is a **stationary Gaussian process**. Equivalently, consumption growth and gross returns are jointly lognormal.

Interpretation. This assumption is what converts the nonlinear Euler equation into a linear forecasting relation. It also makes the likelihood function easy to write down. But it is a **strong** assumption, and a rejection of the model can reflect failure of lognormality rather than failure of CRRA utility alone.

2.5 What the paper does and does not model explicitly

- The paper does *not* solve a fully specified production economy with a particular technology shock process.
- Instead, it works directly with the restrictions that equilibrium Euler equations impose on the joint law of consumption growth and returns.
- The lognormality assumption is therefore a reduced-form restriction on the equilibrium data-generating process.

Interpretation. The paper is not estimating a deep production model. It is estimating a **restricted reduced-form time-series representation** implied by a class of consumption-based asset-pricing models.

2.6 Consumption data

- Sample period: **1959:2–1978:12** for the main monthly exercises.
- Source: seasonally adjusted real consumption from **CITIBASE**, divided by monthly population estimates to form **real per-capita consumption**.
- Two measures are used:
 1. **ND**: nondurables,
 2. **NDS**: nondurables plus services.
- Durables are excluded because the service flow from the durable stock is hard to measure.

Interpretation. Using ND or NDS is not innocuous. Each choice implicitly assumes a different separability structure between the measured component and the rest of consumption.

2.7 Return data

- **Aggregate stock return**: value-weighted return on all NYSE stocks.
- **Individual stock returns**: three Dow Jones industrials — American Brands, Exxon, and IBM.

- **Bond return:** 1-month Treasury bill return.
- Stock returns come from **CRSP**; Treasury bill returns come from **Ibbotson and Sinquefeld**.
- Nominal returns are converted to real returns using the implicit price deflator matching the consumption measure.

Interpretation. Aggregate and individual return exercises are not just different datasets. They are, in effect, different maintained models because lognormality for the aggregate return does not generally coexist with joint lognormality of the individual stock returns.

2.8 Timing assumption behind monthly data

The paper emphasizes that if monthly consumption is used, one is implicitly assuming that the representative agent makes consumption decisions at monthly frequency and knows the month-ahead return relevant for those decisions.

Interpretation. This is a real vulnerability. If the true decision period is shorter than one month, or if measured consumption is time-aggregated and misaligned with returns, then even a correct economic model may look false in the data.

3 Models and empirical specifications (all equations + interpretation)

3.1 The main log-linear restriction

Under joint Gaussianity, the conditional expectation of u_{it} satisfies

$$E(u_{it} \mid \psi_{t-1}) = \exp\left(\mu_{i,t-1} + \frac{\sigma_i^2}{2}\right), \quad (7)$$

where ψ_{t-1} is the econometric information set, $\mu_{i,t-1} = E(U_{it} \mid \psi_{t-1})$, and σ_i^2 is the conditional variance of U_{it} .

Combining this with the Euler equation implies

$$\mu_{i,t-1} = -\log \beta - \frac{\sigma_i^2}{2}. \quad (8)$$

Define the innovation-like object

$$V_{it} \equiv U_{it} - \mu_{i,t-1} = \alpha X_t + R_{it} + \log \beta + \frac{\sigma_i^2}{2}. \quad (9)$$

Then

$$E(V_{it} \mid \psi_{t-1}) = 0, \quad (10)$$

and therefore

$$E(R_{it} \mid \psi_{t-1}) = -\alpha E(X_t \mid \psi_{t-1}) - \log \beta - \frac{\sigma_i^2}{2}. \quad (11)$$

Interpretation.

- All *forecastable* movement in returns must come from forecastable consumption growth.
- The slope linking the two is common across assets and equals $-\alpha$, the coefficient of relative risk aversion.
- Assets differ only through constants involving innovation variances.

3.2 Special cases

- **Risk neutrality:** if $\alpha = 0$, then

$$E(R_{it} \mid \psi_{t-1}) = \text{constant},$$

so log returns are serially uncorrelated.

- **Log utility:** if $\alpha = -1$, then the slope coefficients in regressions of returns on instruments must equal the slope coefficients in regressions of consumption growth on the same instruments.

Interpretation. These are sharp and intuitive implications. Risk neutrality kills return predictability; log utility forces one-for-one comovement of predictable returns and predictable consumption growth.

3.3 The model's predictability formula

The paper derives

$$\text{var}(E(R_{it} \mid \psi_{t-1})) = \alpha^2 \text{var}(E(X_t \mid \psi_{t-1})), \quad (12)$$

which implies

$$R_i^2 = \frac{\alpha^2 \text{var}(E(X_t \mid \psi_{t-1}))}{\text{var}(R_{it} \mid \psi_{t-1}) + \alpha^2 \text{var}(E(X_t \mid \psi_{t-1}))}. \quad (13)$$

Interpretation.

- Risk aversion is **necessary** for predictable returns, because if $\alpha = 0$ then $R_i^2 = 0$.
- But risk aversion is not **sufficient**. If consumption growth is a log random walk, then $E(X_t \mid \psi_{t-1})$ is constant and returns are still unpredictable.

3.4 Forecasting equation for consumption growth

To estimate the model, Hansen and Singleton parameterize the predictable component of log consumption growth as

$$E(X_t \mid \psi_{t-1}) = a(L)'Y_{t-1} + \mu_X, \quad (14)$$

where $a(L)$ is a vector lag polynomial of finite order.

Interpretation. The paper does not estimate a structural consumption function. It estimates a **linear forecasting rule** for consumption growth using lagged consumption growth and lagged returns.

3.5 Restricted VAR representation

Combining the forecasting equation with the return restriction yields a restricted vector autoregressive form,

$$A_0 Y_t = A_1(L) Y_{t-1} + \mu + V_t. \quad (15)$$

The matrix A_0 embeds the common loading α of every return equation on current consumption growth.

Interpretation. This is the econometric heart of the paper. The CRRA consumption-based model becomes a system in which the same preference parameter α restricts all return equations simultaneously.

3.6 Maximum-likelihood estimation

Let θ collect

- α and β ,
- the coefficients in $a(L)$,
- the covariance matrix of the innovation vector V_t .

The conditional log-likelihood has the Gaussian form

$$\ell(\theta) = -\frac{T}{2} \log |\Sigma| - \frac{1}{2} \sum_{t=1}^T \varepsilon_t(\theta)' \Sigma^{-1} \varepsilon_t(\theta), \quad (16)$$

where

$$\varepsilon_t(\theta) = A_0 Y_t - A_1(L) Y_{t-1} - \mu. \quad (17)$$

Interpretation. The paper estimates the free parameters by Gaussian ML and then uses **likelihood-ratio tests** to evaluate the overidentifying restrictions implied by the model.

3.7 Lag length and quarterly comparison

The main monthly models use

$$\text{NLAG} = 2, 4, 6.$$

The paper also estimates two quarterly versions using aggregated nondurable consumption.

Interpretation. This is both a robustness check and a way to see how much the results depend on the time aggregation of consumption.

3.8 A second class of tests using only nominal returns

To avoid direct reliance on measured consumption and the deflator, the paper considers the generalized utility specification

$$U(c_t, \lambda_t) = \frac{c_t^\gamma \lambda_t}{\gamma}, \quad (18)$$

where λ_t is a multiplicative preference shock. Then the Euler equation becomes

$$E_t \left[\beta \left(\frac{c_{t+1}}{c_t} \right)^\alpha \left(\frac{\lambda_{t+1}}{\lambda_t} \right) r_{i,t+1} \right] = 1. \quad (19)$$

Under joint Gaussianity, the difference between the log returns on any two assets satisfies

$$R_{it} - R_{jt} = \frac{\sigma_j^2 - \sigma_i^2}{2} + V_{it} - V_{jt}. \quad (20)$$

Hence the predictable part of a log return spread must be zero.

Interpretation. These tests ask whether **nominal return spreads** can be forecast from lagged information. If they can, the CRRA–lognormal restrictions fail even after allowing for multiplicative preference shocks and avoiding direct consumption measurement.

4 Tables and empirical results

4.1 Table 1: value-weighted aggregate stock return

Read.

- The paper estimates six main monthly models for the value-weighted NYSE return:
 - ND consumption, NLAG = 2, 4, 6;
 - NDS consumption, NLAG = 2, 4, 6.
- For **ND**, the point estimates of α are

$$-0.325, \quad -0.831, \quad -1.25,$$

with β estimates

$$0.9976, \quad 0.9985, \quad 0.9993.$$

- For **NDS**, the point estimates of α are

$$0.359, \quad -0.264, \quad -1.509,$$

with β estimates

$$0.9965, \quad 0.9979, \quad 1.0007.$$

- The unrestricted consumption-growth R_X^2 values range from

0.119 to 0.246,

showing that monthly consumption growth is forecastable.

- The unrestricted return R_R^2 values are much smaller,

0.012 to 0.056,

so stock returns are only weakly forecastable.

- The likelihood-ratio statistics are modest:

6.088, 8.426, 10.622, 4.980, 6.687, 10.932,

with upper-tail p -values between about 0.11 and 0.48.

- Two quarterly models are also reported. Their α estimates are -2.721 and -2.671 , but the standard errors are enormous.

Economic message. At the level of an **aggregate stock index**, the CRRA-lognormal restrictions are not decisively rejected. The model can rationalize some aggregate return predictability with reasonable values of risk aversion. But the price is that the estimates of risk aversion are very noisy and depend a lot on lag length and consumption definition.

TABLE 1
SUMMARY OF MAXIMUM LIKELIHOOD RESULTS FOR
VALUE-WEIGHTED AGGREGATE RETURN

Model	$\hat{\alpha}^*$	$\hat{\beta}^*$	CONS	NLAG	R_C^2	R_R^2	χ^2 [†]	df
1	-.325 (.828)	.9976 (.0032)	ND	2	.187 (.893)	.020 (.893)	6.088	3
2	-.831 (.746)	.9985 (.0030)	ND	4	.206 (.703)	.040 (.703)	8.426	7
3	-1.25 (.647)	.9993 (.0030)	ND	6	.246 (.524)	.056 (.524)	10.622	11
4	.359 (1.880)	.9965 (.0048)	NDS	2	.119 (.827)	.012 (.827)	4.980	3
5	-.264 (1.835)	.9979 (.0045)	NDS	4	.128 (.538)	.028 (.538)	6.687	7
6	-1.509 (1.571)	1.0007 (.0042)	NDS	6	.149 (.551)	.048 (.551)	10.932	11
7‡	-2.721 (3.187)	.9957 (.0627)	ND	2	.201 (.838)	.074 (.838)	5.130	3
8‡	-2.671 (9.306)	.9989 (.1442)	ND	4	.445 (.813)	.111 (.813)	10.030	7

* Standard errors are indicated in parentheses.

† Probability values are indicated in parentheses.

‡ Estimated using quarterly data: 1954:4–1978:4.

TABLE 2
MAXIMUM LIKELIHOOD ESTIMATES FOR MODEL 6
(C = Nondurables + Services, NLAG = 6)

Restricted model ($\hat{\alpha} = -1.5088$ [1.571]; $\hat{\beta} = 1.007$ [.0042]):									
$X =$	.0027	-.334X ₋₁	-.147X ₋₂	+.074X ₋₃	+.086X ₋₄				
	(.0009)	(.071)	(.084)	(.077)	(.074)				
		-.025X ₋₅	+.006X ₋₆	+.002R ₋₁	+.012R ₋₂				
		(.081)	(.069)	(.007)	(.008)				
		+.012R ₋₃	+.004R ₋₄	+.003R ₋₅	-.014R ₋₆				
		(.007)	(.007)	(.008)	(.008)				
Unrestricted model:									
$R =$	.0065	-.502X ₋₁	+.033X ₋₂	-.468X ₋₃	-.623X ₋₄				
	(.005)	(.683)	(.717)	(.722)	(.717)				
		-.827X ₋₅	-.239X ₋₆	+.098R ₋₁	-.056R ₋₂				
		(.705)	(.667)	(.069)	(.069)				
		+.114R ₋₃	+.073R ₋₄	+.076R ₋₅	-.114R ₋₆	$R^2 = .048$			
		(.071)	(.071)	(.071)	(.071)				
$X =$	.0028	-.334X ₋₁	-.144X ₋₂	+.069X ₋₃	+.078X ₋₄				
	(.0005)	(.069)	(.072)	(.073)	(.072)				
		-.033	+.003X ₋₆	+.003R ₋₁	+.011R ₋₂				
		(.071)	(.067)	(.007)	(.007)				
		+.012R ₋₃	+.004R ₋₄	+.004R ₋₅	-.015R ₋₆	$R^2 = .149$			
		(.007)	(.007)	(.007)	(.007)				

NOTE.—Standard errors are indicated in parentheses.

4.2 Table 2: why increasing lag length does not change much

Read.

- Table 2 reports the restricted and unrestricted estimates for **model 6** (NDS, NLAG = 6).
- In the restricted equation for consumption growth, the first two own lags are the important ones:

$$X_t \approx 0.0027 - 0.334X_{t-1} - 0.147X_{t-2} + \dots,$$

while the lagged return coefficients are tiny.

- In the unrestricted system, the X_t equation looks very similar to the restricted one, but the return equation has very low explanatory power:

$$R_R^2 = 0.048, \quad R_X^2 = 0.149.$$

Economic message. Extra lags beyond the first couple of months add little to forecasting consumption. So when NLAG rises, the test gets more overidentified without gaining much fit. That is why the aggregate-stock likelihood-ratio statistics do not explode as lag length increases.

4.3 Table 3: individual Dow Jones returns

Read.

- The paper next uses three individual stock returns: **American Brands, Exxon, and IBM.**
- For ND consumption, the reported estimates are:
 - NLAG = 2: $\alpha = -0.466$, $\beta = 0.995$, $\chi^2 = 310.3$, $df = 25$,
 - NLAG = 4: $\alpha = -1.738$, $\beta = 0.997$, $\chi^2 = 334.7$, $df = 49$.
- For NDS consumption:
 - NLAG = 2: $\alpha = -0.507$, $\beta = 0.995$, $\chi^2 = 238.5$, $df = 25$,
 - NLAG = 4: $\alpha = -4.106$, $\beta = 1.003$, $\chi^2 = 393.7$, $df = 49$.
- The associated upper-tail p -values are essentially zero.

Economic message. The model fails badly once we leave the aggregate stock index and look at **individual equities**. This suggests that any success in the aggregate-stock exercise may rely on aggregation washing out violations that are very visible in disaggregated returns.

TABLE 3
LIKELIHOOD RATIO TESTS FOR THE MODELS OF INDIVIDUAL DOW JONES RETURNS
(1959:2–1978:12)

	$\hat{\alpha}$	$\hat{\beta}$	χ^2	df	Probability
C = Nondurables:					
NLAG = 2	–.466	.995	310.3	25	1.000
NLAG = 4	–1.738	.997	334.7	49	1.000
C = Nondurables plus services:					
NLAG = 2	–.507	.995	238.5	25	1.000
NLAG = 4	–4.106	1.003	393.7	49	1.000

TABLE 4
SUMMARY OF MAXIMUM LIKELIHOOD RESULTS FOR NOMINAL RISK-FREE RETURN

Model	$\hat{\alpha}^*$	$\hat{\beta}^*$	CONS	NLAG	R_c^2	R_R^2	$\chi^{2\dagger}$	df
1	–.164 (.056)	.9997 (.0002)	ND	2	.218	.130	27.34 (.9999)	3
2	–.188 (.060)	.9998 (.0002)	ND	4	.212	.152	33.48 (.9999)	7
3	–.931 (.044)	1.0015 (.0004)	NDS	2	.128	.181	30.08 (.9999)	3
4	–1.289 (.088)	1.0022 (.0006)	NDS	4	.131	.198	30.82 (.9999)	7

* Standard errors are indicated in parentheses.
† Probability values are indicated in parentheses.

4.4 Table 4: 1-month Treasury bill return

Read.

- For Treasury bills, the point estimates of α are

$$-0.164, -0.188, -0.931, -1.289,$$

TABLE 5
SUMMARY OF MAXIMUM LIKELIHOOD RESULTS FOR NOMINAL
RISK-FREE AND VALUE-WEIGHTED RETURNS

Model	$\hat{\alpha}^*$	$\hat{\beta}^*$	CONS	NLAG	χ^2 [†]	df
1	-30.58 (34.06)	1.001 (.0462)	ND	0	Just identified	Just identified
2	-.205	.999	ND	4	170.25 (.9999)	24
3	-58.25 (66.57)	1.088 (.0687)	NDS	0	Just identified	Just identified
4	-.209	1.000	NDS	4	366.22 (.9999)	24

* Standard errors are indicated in parentheses.
† Probability values are indicated in parentheses.

while β stays extremely close to one.

- The return R_R^2 values are now much larger than for aggregate stocks:

$$0.130, 0.152, 0.181, 0.198.$$

- But the likelihood-ratio statistics are

$$27.34, 33.48, 30.08, 30.82,$$

with $df = 3$ or 7 , so the model is strongly rejected.

Economic message. Treasury bill returns are more forecastable than the aggregate stock index, which is not itself a problem for the model. The problem is that the **particular way** their predictability is tied to consumption growth by CRRA–lognormal restrictions is wrong in the data.

4.5 Table 5: stock and bond returns jointly

Read.

- The paper then jointly uses the value-weighted stock return, the Treasury bill return, and nondurable consumption.
- When $NLAG = 0$, the system is just identified and the estimated α is absurdly large in magnitude:

$$-30.58 \text{ for ND, } \quad -58.25 \text{ for NDS,}$$

both with huge standard errors.

- When serial correlation is allowed ($NLAG = 4$), the point estimates collapse to

$$-0.205 \text{ and } -0.209,$$

but the overidentifying restrictions are crushed:

$$\chi^2 = 170.25 \text{ and } 366.22, \quad df = 24.$$

Economic message. This table is extremely important. If one ignores serial correlation and focuses only on unconditional moments, one can back out huge and implausible risk-aversion estimates.

Once time-series predictability is taken seriously, the implied risk aversion becomes much smaller — but the model is then strongly rejected.

4.6 Nominal-return-spread tests

Read.

- Using only nominal returns and no direct consumption data, the paper tests whether return spreads are forecastable.
- For the model with the Treasury bill and the value-weighted stock return,

$$\chi^2(5) = 16.56,$$

with an upper-tail p -value of about 0.005.

- For the model with the Treasury bill and the three individual Dow returns,

$$\chi^2(27) = 53.19,$$

with an upper-tail p -value of about 0.002.

Economic message. These rejections are especially damaging because they do *not* rely on measured real consumption or on the implicit price deflator. The failure of the model is therefore not just a consumption-measurement story.

5 Identification and what is really being tested

5.1 Where identification of α and β comes from

- β is pinned down by the constant term in the log-linear Euler restriction.
- α is pinned down by the **common slope** linking predictable returns to predictable consumption growth across assets and across time.
- Identification therefore uses the whole serial-correlation structure, not just unconditional means.

Interpretation. Table 5 makes this point vividly. With no time-series dynamics (NLAG = 0), the data can imply huge risk-aversion estimates. Once time-series restrictions are imposed, the estimates move dramatically.

5.2 What a rejection means

A rejection of the likelihood-ratio restrictions is a joint rejection of:

- representative-agent pricing,
- time-separable CRRA preferences,

- the chosen separability assumption for ND or NDS,
- stationary Gaussian / lognormal dynamics,
- the timing convention linking monthly consumption to monthly returns,
- and the specific finite-lag forecasting representation.

Interpretation. The tests are powerful but not diagnostic. They tell us the *whole maintained structure* is wrong; they do not isolate which assumption is to blame.

5.3 Why the aggregate stock return looks better than individual stocks

- Aggregation can wash out idiosyncratic nonnormalities and micro-level misspecification.
- The aggregate value-weighted return may be closer to the payoff on “the market” in a representative-agent model than a handful of individual securities.
- But the paper itself notes that assuming lognormality for the aggregate return and for individual stock returns amounts to testing different maintained models.

Interpretation. The partial success of the aggregate-stock exercise should not be read as strong confirmation. It may just mean that aggregation hides the dimensions along which the model fails.

5.4 Timing and measurement concerns

- Monthly consumption data are partly interpolated from lower-frequency benchmark information.
- If the true decision interval is shorter than one month, measured monthly consumption is time-aggregated and noisy relative to the model’s object.
- If the return known to agents when choosing consumption is not aligned with the measured calendar-month return, the empirical system is misspecified.

Interpretation. These are not small technicalities. The paper is unusually honest in emphasizing that the monthly-data exercise may be misspecified even if the underlying theory is right.

5.5 How to read the α estimates economically

- Remember that the coefficient of relative risk aversion is $-\alpha$.
- So $\alpha = -1$ corresponds to relative risk aversion equal to one.
- The most plausible monthly aggregate-stock estimates, especially the longer-lag ones, are roughly in the neighborhood of

$$-\alpha \approx 1 \text{ to } 1.5.$$

- But the standard errors are wide, and one monthly specification even produces $\alpha > 0$, which implies nonconcave utility and is economically implausible.

Interpretation. The point estimates are not crazy for the aggregate stock return. The problem is that they are unstable and the stronger tests reject the model anyway.

5.6 Why the Treasury-bill rejection matters

- Treasury bills are low-variance assets, so the model predicts that if consumption growth is forecastable, T-bill returns may also exhibit noticeable forecastability.
- The data do show larger return R^2 values for Treasury bills than for the aggregate stock return.
- Yet the cross-equation restrictions tying that forecastability to consumption growth are strongly rejected.

Interpretation. This is useful evidence because it shows that the model is failing in the *structure* of return predictability, not simply because stock returns are too noisy.

6 Short summary

- Hansen and Singleton start from a representative-agent CRRA Euler equation and add a joint lognormality assumption for consumption growth and returns.
- Those assumptions imply that predictable log returns are proportional to predictable log consumption growth, with proportionality factor equal to relative risk aversion.
- The model can be estimated as a restricted Gaussian likelihood system.
- Using monthly U.S. postwar data, the aggregate value-weighted stock return models are not strongly rejected and imply economically plausible but noisy estimates of risk aversion.
- The same structure is strongly rejected for individual stock returns, for Treasury bills, and for nominal-return-spread tests that avoid direct consumption measurement.
- The paper therefore provides only limited support for the basic CRRA–lognormal consumption-based model and motivates later work with more general preferences and less parametric estimation methods.

7 Conclusion

7.1 Core conclusion

This paper shows that a representative-agent consumption-based model with time-separable CRRA utility and joint lognormality puts sharp restrictions on return predictability. Those restrictions can be taken directly to the data through a likelihood-based time-series system.

7.2 Economic intuition

The key insight is simple: if investors dislike fluctuations in consumption, then any *forecastable* movement in returns must line up with forecastable movement in consumption growth. The stronger the risk aversion, the stronger that linkage should be.

7.3 Takeaway

The aggregate stock market gives the model some room to survive, but richer tests do not. The paper is therefore best read as both an early empirical application of consumption-based asset pricing and an early demonstration that the standard CRRA–lognormal specification is too narrow to explain the full time-series behavior of returns.